# *Objective Head-Related Transfer Function Evaluation Metrics for More Consistent Perceptual Assessment*

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# Abstract

Head-Related Transfer Functions (HRTFs) are the cornerstone of binaural spatial audio rendering, yet their objective evaluation remains fundamentally flawed. The industry-standard metric, Log-Spectral Distortion (LSD), operates exclusively in the magnitude domain and has been shown to exhibit poor correlation with subjective perceptual quality, particularly when phase-altering processing such as minimum-phase reconstruction is applied. This limitation undermines the reliability of current HRTF personalization and compression pipelines that rely on LSD as their primary quality criterion. This thesis investigates the failure modes of LSD and proposes a Composite Perceptual Metric (CPM) that integrates four complementary objective measures: LSD, Interaural Level Difference (ILD) error, group delay error, and Interaural Time Difference (ITD) error. Using the RIEC HRTF Database from Tohoku University (20 subjects, 865 spatial positions per subject, 512-sample impulse responses at 48 kHz), five degradation conditions were applied to isolate specific metric blindspots: minimum-phase reconstruction, 1/3-octave and 1/1-octave magnitude smoothing, and 12-bit and 8-bit quantization. The central finding confirms the hypothesized "LSD blindspot": the minimum-phase condition achieves the lowest LSD of all conditions (0.030 dB), yet produces the highest CPM score (14.094) due to a group delay error of 2.335 ms—over 10× larger than any other condition. This result provides direct empirical support for the Andreopoulou and Katz (2022) hypothesis that magnitude-only metrics are fundamentally inadequate for HRTF quality prediction. The proposed CPM framework, by incorporating phase-sensitive and binaural cue-based components, captures both magnitude-domain and phase-domain degradations within a single scalar score, offering a more reliable proxy for perceptual HRTF quality assessment. Future work will validate the CPM against subjective listening tests using MUSHRA methodology to establish its correlation with human perception.



**Keywords:** HRTF; Spectral Distortion; Perceptual Evaluation; Spatial Audio; Group Delay; Composite Metric

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# Contents

# List of Figures

# List of Tables

# Definition of Acronyms

| Acronym | Definition |
|---|---|
| HRTF | Head-Related Transfer Function |
| HRIR | Head-Related Impulse Response |
| LSD | Log-Spectral Distortion |
| SD | Spectral Distortion |
| ITD | Interaural Time Difference |
| ILD | Interaural Level Difference |
| CPM | Composite Perceptual Metric |
| GD | Group Delay |
| SOFA | Spatially Oriented Format for Acoustics |
| MUSHRA | MUlti Stimulus test with Hidden Reference and Anchor |
| AMT | Auditory Modeling Toolbox |
| PESD | Perceptually Enhanced Spectral Distance |
| VAD | Virtual Auditory Display |

# Definition of Notations

| **Symbol** | **Meaning** |
|---|---|
| $H(f)$ | Head-Related Transfer Function in the frequency domain |
| $h(t)$ | Head-Related Impulse Response in the time domain |
| $\tau$ | Interaural Time Difference (ITD) |
| $\Delta L$ | Interaural Level Difference (ILD) |
| $\tau_g(f)$ | Group delay as a function of frequency |
| $W_{\text{LSD}}$ | CPM weight for Log-Spectral Distortion (0.20) |
| $W_{\text{ILD}}$ | CPM weight for ILD error (0.25) |
| $W_{\text{GD}}$ | CPM weight for group delay error (0.35) |
| $W_{\text{ITD}}$ | CPM weight for ITD error (0.20) |
| $F_s$ | Sampling frequency (48 000 Hz) |

# 1 Introduction

## 1.1 Overview of Spatial Audio and HRTF

The fundamental goal of virtual auditory display (VAD) and spatial audio systems is to generate a perceptually realistic and immersive three-dimensional auditory experience [1]. The enabling technology for this simulation, particularly over headphones, is the Head-Related Transfer Function (HRTF). The HRTF is a complex, high-dimensional filter that describes the acoustic transformations (diffraction, reflection, and shadowing) imposed by an individual's unique anatomy—primarily the torso, head, and pinnae—on a sound wave before it reaches the eardrum [1].

The profound challenge in spatial audio rendering lies in the high degree of personalization these cues require. HRTFs are highly idiosyncratic; using a non-individualized (or "generic") HRTF, such as one measured on a manikin, has been shown to degrade the spatial experience, leading to localization errors (especially in elevation), front-back confusion, and a failure of externalization, where sound is perceived as being "inside the head" [1]. This has established a critical need for methods to select, personalize, or process HRTFs to better match a given listener [2].

This need for selection and processing, in turn, necessitates a reliable method of evaluation. How is an "appropriate" or "high-quality" HRTF defined? The field has bifurcated into two distinct evaluation paradigms:

**Subjective Assessment (The "Ground Truth"):** This paradigm relies on human listeners to evaluate the perceptual quality of a binaural rendering. As the "ground truth," these methods are perceptually valid but are notoriously time-consuming, expensive, and subject to listener variability [3]. Methodologies are diverse, including:

**Localization Pointing Tasks:** Listeners identify the perceived location of static or dynamic sound sources, and performance is measured by localization error (azimuthal, polar, great-circle) and front-back confusion rates [4].

**Trajectory Rating:** An "indirect" method where listeners rate the "spatial accuracy" or "quality" of a moving sound source (e.g., a trajectory) against a description of its intended path. This is often used as a measure of global spatial quality [2].

**Comparative Quality Assessment:** Listeners rate attributes like timbral coloration, externalization, and overall preference, often using standardized methods like MUSHRA (MUlti Stimulus test with Hidden Reference and Anchor) [5].

**Objective Metrics (The "Proxy"):** This paradigm uses computational, signal-based metrics to quantify the difference between a "reference" (e.g., an individual's measured HRTF) and a "test" (e.g., a processed or personalized HRTF). These methods are fast, perfectly repeatable, and computationally inexpensive. The most ubiquitous metric in HRTF research and development is Spectral Distortion (SD), which measures the RMS difference of magnitude spectra between two HRTFs [6]. Its logarithmic variant, Log-Spectral Distortion (LSD), operates on log-magnitude spectra and better matches the ear's logarithmic intensity perception. Throughout this thesis, "LSD" refers specifically to the log-magnitude variant (Equation 2), while "SD" is used as the general umbrella term encompassing both formulations. Both metrics share the same fundamental limitation: they are computed exclusively in the magnitude domain, averaged across frequency and spatial directions [6].

This thesis addresses the central conflict that defines the research topic: the consistent and well-documented poor correlation between these two paradigms [4]. The objective metrics that the audio engineering community relies on for system design, data reduction, and optimization are fundamentally unreliable predictors of the subjective, perceptual outcome. This disconnect forms the primary obstacle to developing truly effective HRTF personalization and processing techniques.

## 1.2 Motivation

The most direct and compelling evidence of this disconnect is found in the 2022 study "Perceptual Impact on Localization Quality Evaluations of Common Pre-Processing for Non-Individual Head-Related Transfer Functions" by Andreopoulou and Katz [2]. This study was explicitly designed to test the foundational hypothesis that underpins decades of HRTF processing, simplification, and dimensionality reduction.

This hypothesis, as stated by the authors, is that "user-assessments should remain roughly unchanged, as long as the range of signal variations between processed and unprocessed (reference) HRTFs lies within ranges previously reported as perceptually insignificant"

[2]. This assumption is the explicit justification for common techniques like spectral smoothing, Principal Component Analysis (PCA), and filter modeling (e.g., IIR), which all aim to reduce HRTF data complexity by discarding "unnecessary" spectral detail [2]. The "acceptability" of such processing has long been defined by achieving an SD metric below a given threshold, such as 1 or 2 dB [2].

The study's methodology involved having expert listeners subjectively rate the spatial accuracy of moving sound trajectories rendered with different HRTFs. They compared ratings for the original, full-phase HRTFs against ratings for two processed versions: a minimum-phase (Min Ph.) approximation and an Infinite-Impulse Response (IIR) model.

The conclusion of the study is an unambiguous and profound rejection of the foundational hypothesis [2]. The authors found that spectral variations, even those well within "acceptable" objective ranges, can "lead to inversions in quality assessment, resulting in the perceptual rejection of HRTFs that were previously characterized... as the 'most appropriate' or alternatively in the preference of datasets that were previously dismissed as 'unfit' " [2].

This finding is not merely an incremental adjustment but a fundamental invalidation of the field's guiding principle for data reduction. It demonstrates that the industry-standard objective metric (SD) is not just slightly inaccurate, but is actively misleading. The study's logic is inescapable:

**The Premise:** HRTF simplification is based on the idea that much of the spectral detail in a measured HRTF is perceptually "unnecessary" [2].

**The Operational Definition:** "Unnecessary" has been operationally defined by objective metrics. If the SD between the original and simplified HRTF is below an established threshold (e.g., $< 2$ dB), the simplification is deemed "acceptable" and perceptually irrelevant [2].

**The Test:** Andreopoulou & Katz created a minimum-phase processed dataset that objectively met this standard, exhibiting an SD consistently below 1 dB [2].

**The Result:** Subjective assessments of this "objectively acceptable" dataset were not perceptually equivalent. The subjective quality rankings became highly decorrelated from the original, with a median Pearson's correlation coefficient ($r$) of only $\approx 0.5$ (where 1.0

would be identical) [2].

**The Conclusion:** The objective standard of "acceptability" (SD $<$ 1 dB) is invalid. It permitted a processing pipeline that caused a massive subjective decorrelation. This forces the conclusion that the "unnecessary" spectral and/or phase details that were discarded were, in fact, perceptually critical.

## 1.3 Problem Statement

The industry-standard metrics—Mean Square Error (MSE) and Log-Spectral Distortion (LSD)—are derived from linear system theory and quantify "error" as the global spectral difference between a reference HRTF and a processed approximation, averaging magnitude deviations across all frequency bands and spatial directions [2]. This approach operates on the implicit, and incorrect, assumption that the human auditory system weights all spectral distortions equally. In reality, auditory perception is highly non-linear, binaural, and feature-specific: the brain perceives specific psychoacoustic cues such as ITD for low-frequency azimuth, ILD for lateralization, and sparse, high-frequency spectral notches (5–10 kHz) for elevation and front-back discrimination [1]. Current metrics are blind to this hierarchy of features. Consequently, a processing artifact that obliterates a critical 7 kHz pinna notch may yield a numerically lower, "better" LSD score than a benign 2 dB gain fluctuation in a non-critical band.

This metric failure manifests as a "perceptual cliff": minute objective errors—often well within accepted thresholds of "perceptual transparency" (e.g., SD $<$ 1 dB)—can precipitate a catastrophic collapse in subjective quality. Andreopoulou and Katz [2] demonstrated that minimum-phase approximations, despite achieving spectrally "perfect" LSD scores, resulted in a near-total decorrelation of subjective rankings ($r \approx 0.5$). The minimum-phase simplification preserved the *magnitude* spectrum (satisfying LSD) but destroyed the temporal fine structure and group delay cues essential for localization and externalization. This creates a systemic risk: algorithms are being trained to minimize a loss function (LSD) that does not correlate with perceptual realism [7].

Furthermore, the prevalent evaluation paradigm is fundamentally monaural, assessing each ear in isolation. This architecture ignores the binaural nature of spatial hearing, where the *difference* between the ears is often more significant than the absolute signal at either ear [3]. A 1 dB boost in the left ear and a 1 dB cut in the right ear might average out to a low monaural error, but they combine to create a 2 dB ILD shift—a perceptually significant change in perceived source location.

The consequences of such "false positive" ratings are compounded by "false negatives," where perceptually beneficial modifications are penalized. Because LSD penalizes *any* deviation from the reference equally, it cannot distinguish between distortion that destroys a cue and modification that enhances it [8]. This indiscriminacy forces researchers to rely on expensive subjective listening tests as the only reliable ground truth, creating a bottleneck that stifles the iterative design of personalization algorithms [4]. The problem is further exacerbated by the increasing reliance on machine learning for HRTF generation: when deep learning models are trained using MSE or LSD as the loss function, they are incentivized to produce spectrally smooth, "average" HRTFs that lack the sharp, idiosyncratic spectral features necessary for individual externalization [7].

## 1.4 Objectives and Scope

This research aims to analyze the specific failure modes of current metrics and propose a framework for perceptually-informed evaluation. The specific objectives are:

1. **Demonstrate the LSD blindspot:** Empirically confirm that LSD fails to detect phase-domain distortions by applying controlled degradation conditions—particularly minimum-phase reconstruction—to a large HRTF database.

2. **Develop phase-sensitive metrics:** Introduce group delay error and ITD error as complementary metrics that capture the phase and temporal distortions that LSD misses.

3. **Propose a Composite Perceptual Metric (CPM):** Integrate magnitude-domain (LSD), binaural (ILD error), and phase-domain (group delay error, ITD error) components into a single weighted scalar score.

4. **Validate the composite framework:** Demonstrate that the CPM correctly identifies degradation conditions that LSD misjudges, while maintaining sensitivity to magnitude-domain distortions.

The scope of this thesis is restricted to objective analysis. Three fundamental limitations of SD/LSD motivate the proposed metrics:

SD and LSD are, by definition, calculated on the magnitude (or log-magnitude) spectra [6]. They are fundamentally blind to all phase information. This is a critical, and often fatal, flaw. The long-held assumption that the human auditory system is "phase-deaf" is a dangerous oversimplification, especially in the context of HRTFs [9]. While less sensitive to absolute phase, the relative phase between the ears forms the Interaural Time Difference (ITD), which is the dominant psychoacoustic cue for localizing sounds in the horizontal plane, particularly at low frequencies [8]. Furthermore, the minimum-phase approximation itself, while a common simplification, is not perceptually neutral and can lead to inferior localization performance for directions near the interaural axis [10]. The Andreopoulou & Katz study [2] provides the ultimate demonstration: by only modifying phase-related components (and introducing co-incident spectral smoothing), subjective perception was radically altered. The magnitude-only SD metric, by its very design, was incapable of detecting this critical perceptual change.

SD/LSD is typically calculated as an average of two monaural comparisons: the reference left-ear HRTF is compared to the processed left-ear HRTF, and the reference right-ear is compared to the processed right-ear [6]. This architecture completely misses the binaural nature of spatial hearing. Spatial perception arises from the central nervous system's comparison and difference between the signals arriving at both ears [8]. A metric that is purely monaural cannot capture errors in these critical binaural cues. For example, a subtle, anti-correlated error—a 1 dB boost in the left ear and a 1 dB cut in the right ear at the same frequency—could result in a low (good) average SD score, but would create a massive, 2 dB error in the Interaural Level Difference (ILD), which is perceptually disruptive [9].

Finally, SD/LSD (and its root, MSE) is a globally-averaged metric. It treats errors in all frequency bins (or log-bins) with relatively equal importance. Human perception is profoundly non-uniform. Psychoacoustic research has unequivocally shown that localization in the median (sagittal) plane—distinguishing front from back, and up from down—is heavily dependent on specific, high-frequency spectral features (notches and peaks) created by pinna reflections [8]. A globally-averaged metric like SD will "drown out" a critical, perceptually devastating error (e.g., shifting a single 8 kHz pinna notch by 500 Hz) with "good" performance across other frequencies [3].

## 1.5 Thesis Structure

The preceding sections established the core conflict in HRTF evaluation and detailed the fundamental flaws of the ubiquitous Spectral Distortion metric. The remaining chapters of this thesis are organized as follows. Chapter 2 reviews the literature on HRTF evaluation, deconstructing the mechanisms of objective metric failure using detailed case studies on HRTF processing techniques and surveying the modern alternatives that have been proposed. Chapter 3 presents the methodology of the experimental investigation, describing the dataset, degradation conditions, proposed metrics, and evaluation strategy. Chapter 4 details the implementation, presents the experimental results, and discusses their implications. Finally, Chapter 5 summarizes the contributions and outlines directions for future work, including subjective validation via MUSHRA listening tests.

# 2 Literature Review

## 2.1 Introduction

To understand how to build better metrics, it is essential to deconstruct the mechanisms of the failure demonstrated in the Andreopoulou & Katz [2] case study. The minimum-phase and IIR modeling conditions serve as perfect, isolated examples of how specific processing techniques expose the flaws of magnitude-only metrics.

**The Minimum-Phase Fallacy: Isolating the Failure of Magnitude-Only Metrics**

The "minimum-phase plus delay" model has long been a cornerstone of "perceptually-based" HRTF processing [2]. This model is based on seminal studies, such as Kistler & Wightman (1992) [11], which argued that an HRTF's phase response can be adequately modeled as a minimum-phase component (derived from its magnitude spectrum) plus a simple frequency-independent time delay (to account for the ITD) [11].

The Andreopoulou & Katz [2] study implemented this by decomposing the 256-tap full-phase HRTFs using a Hilbert transform and then truncating them to 64-tap filters. This truncation inherently introduced a degree of spectral smoothing. Critically, the study's authors then re-inserted the original, position-dependent ITD values to ensure ITD was consistent across all conditions [2].

The results of this experiment are the most damning evidence against SD as a perceptual metric:

**Objective:** The SD between the full-phase and minimum-phase HRTFs was consistently below 1 dB [2], well within the range universally considered "perceptually insignificant."

**Subjective:** The subjective quality rankings collapsed. The median correlation ($r$) between the full-phase and minimum-phase ratings was $\approx 0.5$ [2].

This was not just statistical "noise." The analysis of the rating categories revealed that $\approx 18\%$ of HRTFs rated as "High" quality in the full-phase condition were rated as "Low" quality in the minimum-phase condition [2]. This is a complete inversion of perceptual assessment. Because the ITD—the primary horizontal localization cue—was controlled for [2], the massive subjective decorrelation cannot be blamed on incorrect horizontal positioning.

The failure must, therefore, stem from the two "subtle" forms of information that were discarded by the processing:

**True Mixed-Phase Information:** The original HRTFs are mixed-phase. The minimum-phase approximation discards all non-minimum-phase components. The study proves this information is not perceptually irrelevant.

**Spectral Fine-Structure:** The truncation resulted in spectral smoothing. While this produced a low globally-averaged SD, it may have been "perceptually catastrophic" by blurring the very sparse, high-frequency pinna notches used for elevation and externalization.

## 2.2 Previous Work

The quest to quantify the similarity between Head-Related Transfer Functions (HRTFs) has evolved from simple signal processing heuristics to complex, data-driven perceptual models. This section reviews the trajectory of HRTF evaluation, highlighting the limitations of traditional metrics and the emergence of modern, perception-based frameworks.

### 2.2.1 The Era of Spectral Distortion and Linear Metrics (1990s – 2010s)

For decades, the dominant paradigm in HRTF evaluation has been the calculation of spectral differences between a reference (measured) HRTF and a test (processed/modeled) HRTF. The most ubiquitous metric is Spectral Distortion (SD), often calculated as the Root Mean Square (RMS) difference of the log-magnitude spectra:

$$\mathrm{SD} = \sqrt{\frac{1}{N}\sum_{n=1}^{N}\left(20\log_{10}|H_{\mathrm{ref}}(f_n)| - 20\log_{10}|H_{\mathrm{test}}(f_n)|\right)^2} \tag{1}$$

This metric's widespread adoption was driven by its computational simplicity and mathematical convexity, making it an ideal candidate for cost functions in optimization algorithms such as spherical harmonic decomposition or Principal Component Analysis (PCA) reconstruction [1].

Early validation attempts focused on establishing "just noticeable difference" (JND) thresholds for SD. Studies in the 1990s and 2000s often cited thresholds (e.g., 1 dB or

2 dB) below which spectral changes were assumed to be inaudible. This led to a pervasive engineering assumption: if a compression algorithm or modeled filter achieves a mean SD lower than the threshold, the processing is "transparent." However, limitations were evident early on. Kistler and Wightman (1992) noted that while PCA could reconstruct HRTFs with low spectral error, the perceptual validity relied heavily on preserving specific principal components that carried localization cues, not just minimizing global variance [11].

Alternative distance metrics attempted to refine SD. The Log-Spectral Distortion (LSD) emphasizes differences in the logarithmic domain, better matching the ear's dynamic range intensity perception. Other variants included the Inter-Subject Spectral Difference (ISSD) and correlation-based metrics [12]. However, a comprehensive 2023 study by Doma, Brožová, and Fels, titled *"Examining the interrelation behavior of distance metrics for head-related transfer function evaluation,"* systematically analyzed seven common metrics (including MSE, Critical-Band MSE, and Mel-frequency Cepstral Distortion). Their analysis, performed on the ITA HRTF database using PCA-based reconstruction errors, revealed a critical redundancy: most metrics were highly correlated with one another, providing little unique information. More importantly, the study concluded that while these metrics could describe signal differences, they exhibited "case-dependency" that made them unreliable predictors of audibility across different types of spectral distortions [12]. They argued that no single linear metric could suffice, suggesting that the complexity of auditory perception required a "higher-level metric" or a combination of descriptors.

### 2.2.2 The Breakdown of the Objective-Subjective Correlation

The pivotal critique of magnitude-only metrics was crystallized in the 2022 study by Andreopoulou and Katz, *"Perceptual Impact on Localization Quality Evaluations of Common Pre-Processing for Non-Individual Head-Related Transfer Functions"* [2]. This work serves as the primary motivation for the current research.

Andreopoulou and Katz designed an experiment explicitly to stress-test the correlation between SD and perceptual spatial quality. They utilized expert listeners to rate the trajectory quality of sound sources rendered with three versions of HRTFs: the original raw measurements, a minimum-phase approximation (re-inserting ITD), and an IIR filter model.

**The Minimum-Phase Failure:** The minimum-phase condition achieved an objective score that was theoretically "perfect" by industry standards, with SD consistently below 1 dB. Yet, subjective results showed a massive decorrelation ($r \approx 0.5$) compared to the reference.

**The Perceptual Inversion:** Approximately 18% of HRTFs rated as "High Quality" in the reference set were downgraded to "Low Quality" in the minimum-phase set, despite the negligible objective error.

**The IIR Cliff:** The IIR models, which introduced slightly higher SD ($> 1.5$ dB), caused a total collapse of perceptual correlation ($r \approx 0.3$), essentially rendering the objective metric useless for prediction.

This study definitively proved that SD is blind to phase distortion and fine spectral structure loss—two factors that are critical for maintaining the stability and plausibility of a virtual sound source [2]. It highlighted that the "unnecessary" details discarded by minimum-phase assumptions are, in fact, perceptually salient.

### 2.2.3 Computational Auditory Models (CAMs) as Metrics

In response to the failure of signal-based metrics, researchers began employing Computational Auditory Models (CAMs)—sophisticated algorithms that simulate the physiological processing of the ear and brain—as evaluation tools. Instead of measuring the distance between signals, these methods measure the distance between the *predicted internal representations* of those signals.

Engel et al. (2022) [13] pioneered the systematic use of the Auditory Modeling Toolbox (AMT) to evaluate HRTF preprocessing methods for Ambisonics rendering. In their study, *"Assessing HRTF preprocessing methods for Ambisonics rendering through perceptual models,"* they replaced human listeners with three specific validated models:

**Reijniers et al. (2014)** [14]: A Bayesian ideal-observer model for sound localization. It predicts localization uncertainty and error by comparing the input binaural signal against a learned template of the subject's HRTF cues.

**Baumgartner et al. (2021)** [15]: A model for sound externalization. This model analyzes monaural spectral cues and interaural deviations to predict the degree to which a sound is perceived as "outside the head" versus "inside the head."

**Jelfs et al. (2011)** [16]: A model for speech intelligibility in spatial noise.

Engel et al. demonstrated that these models could successfully differentiate between preprocessing methods (e.g., MagLS vs. Spline interpolation) even when standard spectral metrics showed ambiguous results [13]. The study validated that the "BiMagLS" method (Bilateral Magnitude Least Squares) preserved localization cues better than traditional approaches. This work established a new methodology: using "virtual listeners" (CAMs) to run thousands of trials that would be impossible with human subjects.

### 2.2.4 Data-Driven and Learned Metrics

A parallel stream of research has moved towards *learning* the perceptual metric directly from data, leveraging machine learning (ML) to bridge the gap between signal and perception.

Ananthabhotla, Ithapu, and Brimijoin (2021) [17] introduced a landmark framework in their JASA Express Letter, *"A framework for designing head-related transfer function distance metrics that capture localization perception."* They argued that the relationship between HRTF spectral features and localization is highly non-linear. They proposed a "Manifold Learning" approach:

**Pre-training:** They first trained a Convolutional Neural Network (CNN) on a large database of HRTFs (123 subjects) to predict the spatial location (azimuth/elevation) of a sound source. This forced the network to learn which spectral features are actually useful for localization (e.g., notches) and which are irrelevant.

**Metric Construction:** The internal representation (latent space) of this neural network was then used as the "metric." The distance between two HRTFs was defined as the Euclidean distance between their embeddings in this learned latent space, rather than their raw spectra.

**Fine-Tuning:** The model was further fine-tuned using sparse perceptual data from human listening tests.

They demonstrated that this "Neural Metric" aligned significantly better with human localization performance than SD/LSD. The metric effectively "learned" that a 1 dB error in a spectral notch is far more important than a 1 dB error in a spectral peak, mirroring human sensitivity [17].

Building on this concept, Zhang et al. (2025) [4] recently proposed *"Towards Perception-Informed Latent HRTF Representations."* They addressed the issue that most Deep Learning models for HRTF generation (e.g., autoencoders) are trained with an MSE loss function, which inherently produces spectrally smooth, "blurry" HRTFs that lack perceptual detail. They introduced a supervision method using Metric Multidimensional Scaling (MMDS). By calculating a "perceptual distance matrix" (using auditory models like Baumgartner's) and forcing the latent space of the Autoencoder to respect these distances, they ensured that the generated HRTFs were perceptually optimized, not just mathematically averaged [4].

### 2.2.5 Recent Advances: PESD and BINAQUAL (2024–2025)

The most recent literature has seen the introduction of specialized metrics designed to replace SD entirely.

Armstrong et al. (2024) [18] published *"Perceptually enhanced spectral distance metric for head-related transfer function quality prediction"* in JASA. This work represents a direct evolution of the standard SD metric. The authors conducted listening tests to identify exactly *why* LSD fails. Based on these data, they engineered the Perceptually Enhanced Spectral Distance (PESD). Unlike a black-box neural network, PESD is an algorithmic metric that incorporates:

**Threshold Discrimination:** Ignoring errors below auditory detection thresholds.

**Feature Combination:** Explicitly weighting spectral peaks and notches differently.

**Binaural Weighting:** Incorporating interaural differences rather than just monaural spectra.

The authors claim PESD exhibits "superior performance in prediction error and correlation with actual perceptual results" compared to all previous spectral metrics [18].

Finally, Panah et al. (2025) [19] introduced BINAQUAL. Adapted from the AMBIQUAL metric for Ambisonics, BINAQUAL is a full-reference metric specifically for binaural audio. It utilizes a "Neuro-SIM" (NSIM) approach, calculating similarity based on phaseograms and structural similarity indices (SSIM) applied to the time-frequency representation of the binaural signal. It explicitly models the preservation of localization cues (ITD/ILD) and has been validated against MUSHRA listening test scores, showing strong correlation for evaluating compression artifacts and spatial degradations [19].

Concurrently, Weckbecker et al. (2025) [20] proposed QASTAnet, a DNN-based quality metric for spatial audio presented at ICASSP 2025. QASTAnet learns to predict subjective spatial quality scores from the time-frequency representation of binaural signals, demonstrating the growing trend toward learned metrics that bypass hand-crafted features. Additionally, Fantini et al. (2025) [21] published a comprehensive survey on machine learning techniques for HRTF individualization in IEEE Open Journal of Signal Processing, which confirms that MSE/LSD remain the dominant objective functions in deep learning HRTF pipelines despite their well-documented perceptual limitations—further motivating the need for composite alternatives like the CPM framework proposed in this thesis.

## 2.3 Research Gaps

Despite these advances, a unified, universally accepted standard is missing.

**Fragmented Landscape:** The field has "localization metrics" (Reijniers [14]), "coloration metrics" (SD/LSD), and "externalization metrics" (Baumgartner [15]), but no single score that captures the holistic "Spatial Quality."

**Implementation Barriers:** Neural metrics [17] require training data and GPU resources; CAMs [13] are computationally heavy. A lightweight, robust metric like PESD needs broader independent validation.

**Phase Blindness:** While BINAQUAL addresses phase, many "enhanced" spectral metrics (like PESD) often still focus heavily on magnitude, potentially missing the "phase-sensitive" distortions highlighted by Andreopoulou & Katz [2].

This thesis aims to bridge these gaps by proposing and evaluating a composite metric framework that explicitly incorporates phase-sensitive and binaural cue-based components alongside the traditional magnitude-domain analysis.

Having established that Spectral Distortion fails (Chapter 1) and why it fails (Chapter 2), the core of the research question—the improvement of objective metrics—must be addressed. The solution is to move away from simple, perceptually-blind signal comparisons and toward metrics that are explicitly designed to measure what humans perceive.

The solutions can be organized into a framework of escalating sophistication:

**Level 1: Measure the Cues.** Stop comparing the entire spectrum and instead compare the specific, known psychoacoustic cues (ITD, ILD) that the brain uses [22].

**Level 2: Model the Percept.** Use validated, computational models of human hearing to predict the subjective outcome (e.g., localization error, externalization) [23].

**Level 3: Learn the Perceptual Space.** Use machine-learning (ML) and data-driven approaches to learn a metric or a data representation that is inherently aligned with human perception [4].

## 2.4 Summary

The preceding sections highlighted a fundamental failure of traditional metrics like Spectral Distortion (SD) to predict subjective HRTF quality. This failure is rooted in SD's inability to account for critical phase information, binaural comparison, and the nonlinear weighting of sparse spectral cues by the human auditory system. The analysis of minimum-phase and IIR modeling revealed a "perceptual cliff," where minute objective errors can cause catastrophic subjective assessment collapses, proving that new, perceptually-anchored metrics are necessary. The following chapters detail the methodology used to empirically demonstrate these failure modes and the proposed composite metric framework designed to address them.

# 3 Methodology

This chapter describes the experimental methodology designed to empirically demonstrate the failure of Log-Spectral Distortion (LSD) and to validate the proposed Composite Perceptual Metric (CPM). Unlike studies that rely on subjective listening tests—which are expensive, time-consuming, and subject to listener variability—this investigation adopts a purely objective approach: deliberately engineering degradation conditions that are known to exploit specific metric blindspots, then evaluating whether the proposed metrics correctly identify the distortions that LSD misses.

## 3.1 Dataset

The experiment utilizes the RIEC HRTF Database, maintained by the Research Institute of Electrical Communication at Tohoku University [24]. This database was selected for several reasons:

- **Subject diversity:** The database contains 51 subjects (40 male, 11 female), providing a broad sampling of anatomical variability in pinna shape, head size, and torso dimensions.
- **Dense spatial sampling:** Each subject's HRTF is measured at 865 spatial positions, providing comprehensive coverage of the sphere surrounding the listener.
- **High temporal resolution:** Head-Related Impulse Responses (HRIRs) are stored as 512-sample filters at a sampling rate of $F_s = 48{,}000$ Hz, yielding a frequency resolution of approximately 93.75 Hz and a Nyquist frequency of 24 kHz.
- **Standard format:** All data are stored in the Spatially Oriented Format for Acoustics (SOFA), enabling interoperability with standard spatial audio toolchains.

From the full database, a subset of 20 subjects was selected: 10 male and 10 female. This balanced selection ensures that any observed metric behavior is not confounded by gender-specific anatomical differences (e.g., smaller pinnae in female subjects, which shift pinna notch frequencies upward). The 20-subject subset yields a total of $20 \times 865 = 17{,}300$ spatial positions per degradation condition, providing robust statistical power for cross-subject averaging.

## 3.2 Degradation Conditions

Five degradation conditions were designed to act as “metric-adversarial attack vectors”—each targets a specific type of signal distortion that is known or hypothesized to expose a blindspot in traditional magnitude-only metrics. The conditions are organized into three categories.

**Condition A: Minimum-Phase Reconstruction.** This condition directly replicates the processing shown to break LSD in the Andreopoulou & Katz study [2]. Each HRIR is decomposed using a real-cepstrum method: the log-magnitude spectrum is computed, inverse-transformed to obtain the cepstrum, and a causal window is applied to extract the minimum-phase component. The original Interaural Time Difference (ITD) is estimated via onset detection (the first sample exceeding 1% of the peak amplitude) and re-inserted by time-shifting the minimum-phase impulse response. This ensures that the ITD cue is preserved while the mixed-phase information and excess group delay structure are destroyed.

*Rationale:* This condition is expected to produce near-zero LSD (since the magnitude spectrum is perfectly preserved by the minimum-phase reconstruction) while generating large group delay errors. It is the primary test of whether the proposed phase-sensitive metrics detect distortions that LSD cannot.

**Condition B: Magnitude Smoothing (1/3-octave and 1/1-octave).** The HRIR magnitude spectrum is smoothed using fractional-octave averaging while preserving the original phase. For each frequency bin $f_k$, the smoothed magnitude is computed as the arithmetic mean of all magnitude values within the band $[f_k/2^{1/(2B)},\ f_k \cdot 2^{1/(2B)}]$, where $B$ is the fractional-octave bandwidth (3 for 1/3-octave, 1 for 1/1-octave). The smoothed magnitude is recombined with the original phase to reconstruct the HRIR.

*Rationale:* Magnitude smoothing simulates the spectral detail loss that occurs in common HRTF compression techniques (e.g., PCA truncation, spherical harmonic decomposition at low orders). The 1/3-octave condition represents mild smoothing (approximately matching critical-band resolution), while the 1/1-octave condition represents aggressive smoothing that is expected to destroy pinna notch features. Both conditions should produce measurable LSD and ILD errors, serving as a “sanity check” that the proposed metrics respond appropriately to magnitude-domain distortions.

**Condition C: Magnitude Quantization (12-bit and 8-bit).** The HRIRs are uniformly quantized to a reduced bit depth. The signal is normalized to its peak amplitude, mapped to $2^b$ discrete levels (where $b$ is the bit depth), and then rescaled.

*Rationale:* Quantization introduces broadband noise that affects both magnitude and phase, but at different severity levels. The 12-bit condition introduces minimal distortion (quantization noise floor approximately $-72$ dB), while the 8-bit condition introduces clearly audible noise ($-48$ dB). These conditions test whether the metrics scale appropriately with increasing distortion severity.

## 3.3 Proposed Metrics

Four individual metrics are computed for each subject-condition pair, plus a composite score. All metrics are computed over the frequency range 200 Hz to 16,000 Hz, which spans the perceptually relevant bandwidth for spatial hearing while avoiding DC artifacts and high-frequency aliasing.

**Metric 1: Log-Spectral Distortion (LSD).** The baseline metric, included as the control against which all other metrics are compared. LSD is defined as:

$$\mathrm{LSD} = \sqrt{\frac{1}{K}\sum_{k=1}^{K}\left(20\log_{10}\frac{|H_{\mathrm{ref}}(f_k)|}{|H_{\mathrm{test}}(f_k)|}\right)^2} \tag{2}$$

where $K$ is the number of frequency bins in the analysis range. LSD is computed independently for each ear at each spatial position, then averaged across ears and positions. A regularization term ($\epsilon = 10^{-12}$) is added to the denominator to prevent division by zero.

**Metric 2: Interaural Level Difference (ILD) Error.** The ILD at each frequency bin is defined as $\mathrm{ILD}(f) = 20\log_{10}(|H_L(f)|/|H_R(f)|)$. The ILD error is computed as the RMS difference between the reference and test ILD spectra:

$$\mathrm{ILD}_{\mathrm{err}} = \sqrt{\frac{1}{K}\sum_{k=1}^{K}\left(\mathrm{ILD}_{\mathrm{ref}}(f_k) - \mathrm{ILD}_{\mathrm{test}}(f_k)\right)^2} \tag{3}$$

This metric is inherently binaural—it measures the *difference between ears* rather than the error at each ear independently. The just noticeable difference (JND) for ILD is approximately 1 dB [9], providing a psychoacoustic anchor for interpreting the metric values.

**Metric 3: Group Delay Error.** Group delay is the negative derivative of the phase response with respect to frequency. It quantifies the time delay experienced by each frequency component of the signal. The group delay is computed using the conjugate-division formulation for numerical stability:

$$\tau_g(f) = -\frac{\mathrm{Re}\left[H'(f) \cdot H^*(f)\right]}{|H(f)|^2 + \epsilon} \tag{4}$$

where $H'(f)$ is the DFT of the time-weighted impulse response $n \cdot h(n)$, $H^*(f)$ is the complex conjugate of $H(f)$, and $\epsilon$ is a regularization constant. The group delay error is computed as the RMS difference (in milliseconds) between the reference and test group delays, averaged across ears and positions.

*This is the key innovation of the proposed framework.* Group delay is sensitive to phase distortion by construction—it is derived from the phase spectrum. Minimum-phase reconstruction, which preserves magnitude perfectly but replaces the original phase with a computed minimum-phase response, is expected to produce large group delay errors while generating near-zero LSD.

**Metric 4: Interaural Time Difference (ITD) Error.** ITD is estimated via cross-correlation between the left and right ear HRIRs. The lag corresponding to the maximum of the cross-correlation function is converted to milliseconds:

$$\mathrm{ITD} = \frac{\arg\max_\ell \sum_n h_L(n) \cdot h_R(n+\ell)}{F_s} \times 1000 \tag{5}$$

The ITD error is the RMSE between the reference and test ITD values across all spatial positions. The JND for ITD is approximately 10–30 $\mu$s [9].

**Composite Perceptual Metric (CPM).** The four metrics are combined into a single scalar score via weighted summation after normalization to psychoacoustically meaningful reference values:

$$\mathrm{CPM} = \frac{W_{\mathrm{LSD}} \cdot \mathrm{LSD}}{\mathrm{LSD}_{\mathrm{ref}}} + \frac{W_{\mathrm{ILD}} \cdot \mathrm{ILD}_{\mathrm{err}}}{\mathrm{ILD}_{\mathrm{ref}}} + \frac{W_{\mathrm{GD}} \cdot \mathrm{GD}_{\mathrm{err}}}{\mathrm{GD}_{\mathrm{ref}}} + \frac{W_{\mathrm{ITD}} \cdot \mathrm{ITD}_{\mathrm{err}}}{\mathrm{ITD}_{\mathrm{ref}}} \tag{6}$$

The normalization references are derived from established psychoacoustic Just Noticeable

Difference (JND) thresholds reported in the literature:

- $\text{LSD}_{\text{ref}} = 2.0$ dB: This value reflects the upper bound of the "perceptually insignificant" SD range widely used in the HRTF literature. Andreopoulou and Katz [2] note that SD values below 1–2 dB have traditionally been considered acceptable; we adopt 2.0 dB as the reference to ensure that only distortions exceeding this established threshold contribute substantially to the composite score.
- $\text{ILD}_{\text{ref}} = 1.0$ dB: The JND for ILD is approximately 0.5–1.5 dB depending on frequency and stimulus type [9]. We adopt 1.0 dB as a representative mid-range value, consistent with the ILD JND reported by Blauert for broadband stimuli.
- $\text{GD}_{\text{ref}} = 0.08$ ms: Group delay JNDs are frequency-dependent and range from approximately 0.05 ms at high frequencies to 0.5 ms at low frequencies [9]. The value of 0.08 ms (approximately 3.8 samples at 48 kHz) represents the lower end of this range, reflecting the high perceptual sensitivity to temporal fine-structure in the 4–12 kHz pinna notch region that is most critical for elevation perception.
- $\text{ITD}_{\text{ref}} = 0.03$ ms: The ITD JND is approximately 10–30 $\mu$s for broadband stimuli presented at the midline [9]. We adopt 0.03 ms (30 $\mu$s) as the upper end of this range, corresponding to approximately 1.4 samples at 48 kHz.

The weights are $W_{\text{LSD}} = 0.20$, $W_{\text{ILD}} = 0.25$, $W_{\text{GD}} = 0.35$, $W_{\text{ITD}} = 0.20$, summing to 1.0. The rationale for the weight assignment follows a principled hierarchy based on two criteria: (1) the *diagnostic gap*—how much information the metric provides beyond what LSD already captures—and (2) the *perceptual salience* of the cue class for spatial hearing:

- **Group delay ($W_{\text{GD}} = 0.35$, highest weight):** This component receives the highest weight because it is the *only* metric in the framework that directly measures phase-domain distortion—the precise failure mode identified by Andreopoulou and Katz [2]. Since the central hypothesis of this thesis is that phase blindness is the primary failure of LSD, the phase-sensitive component must be weighted most heavily to ensure the CPM detects what LSD cannot.
- **ILD error ($W_{\text{ILD}} = 0.25$):** ILD is the dominant lateralization cue above approximately 1.5 kHz [9] and is inherently binaural—it measures the *difference between ears* rather than the absolute error at each ear. This binaural perspective is absent

from LSD, making ILD error the second most diagnostic component.

- **LSD ($W_{\text{LSD}} = 0.20$) and ITD error ($W_{\text{ITD}} = 0.20$):** LSD is included as the baseline magnitude-domain component to ensure the CPM remains sensitive to spectral distortions that *are* captured by traditional metrics. ITD error, while perceptually critical for low-frequency azimuthal localization, is assigned equal weight to LSD because ITD is typically well-preserved by most HRTF processing pipelines (as confirmed by our results in Table 1).

It is important to note that these weights are heuristic, informed by psychoacoustic principles rather than optimized against subjective data. Future work (Section 5.2) describes how these weights can be empirically optimized via regression against subjective MUSHRA scores once such data become available. The normalization by JND-based references ensures that each component contributes to the composite score in proportion to its perceptual significance: a value of 1.0 for any normalized component indicates that the error is at the threshold of detectability.

## 3.4 Evaluation Strategy

The evaluation proceeds in three stages:

**Stage 1: Per-Subject, Per-Condition Metric Computation.** For each of the 20 subjects and each of the 5 degradation conditions, all four metrics (LSD, ILD error, GD error, ITD error) and the CPM composite are computed. This yields a results matrix of dimensions $20 \times 5 \times 5$ (subjects $\times$ conditions $\times$ metrics).

**Stage 2: Cross-Condition Comparison.** The mean and standard deviation of each metric are computed across the 20 subjects for each condition. The key diagnostic question is: *do the individual metrics and the CPM composite correctly rank the degradation conditions in order of expected perceptual severity?* Specifically, the minimum-phase condition should be ranked as severely degraded by the CPM (due to group delay error) despite being ranked as minimally degraded by LSD.

**Stage 3: Blindspot Identification.** The LSD vs. CPM relationship is analyzed to identify conditions where the two metrics diverge. A large divergence (low LSD, high CPM) constitutes empirical evidence of a metric blindspot. The minimum-phase condition is the predicted primary blindspot.

# 4 Implementation, Results, and Discussion

This chapter presents the experimental implementation, reports the quantitative results, and discusses their implications for HRTF evaluation methodology. All code, data, and figures are available in the accompanying repository[1].

## 4.1 Experimental Setup

The experiment was implemented in Python using the following libraries: NumPy for numerical computation, SciPy for signal processing operations, SOFAr for reading SOFA-format HRTF files, Pandas for tabular data management, and Matplotlib for visualization.

The RIEC HRTF Database [24] was organized into `male/` and `female/` subdirectories, each containing SOFA files. The first 10 subjects from each subdirectory were loaded, yielding a balanced set of 20 subjects ($N_{\mathrm{male}} = 10$, $N_{\mathrm{female}} = 10$). Each subject's HRIR array has dimensions $(865, 2, 512)$, corresponding to 865 spatial positions, 2 ears (left/right), and 512 time-domain samples at $F_s = 48{,}000$ Hz. All metric computations were restricted to the frequency range $f \in [200, 16{,}000]$ Hz.

## 4.2 Degradation Conditions

Five degradation conditions were applied to each subject's HRIR array:

**Condition 1: Minimum-Phase Reconstruction.** The real-cepstrum method was employed. For each individual HRIR (at each spatial position and ear), the 512-sample impulse response was zero-padded to 1024 samples, transformed to the frequency domain, and the log-magnitude spectrum was computed. The cepstrum was obtained via inverse FFT, and a causal window (doubling the positive-time cepstral coefficients while zeroing the negative-time coefficients) was applied. The minimum-phase impulse response was recovered by exponentiating the FFT of the windowed cepstrum and taking the inverse FFT. The original ITD was estimated via onset detection (first sample exceeding 1% of peak amplitude) and re-inserted by circularly shifting the earlier-arriving ear's impulse response.

**Condition 2: 1/3-Octave Magnitude Smoothing.** For each HRIR, the magnitude

[1] Source code available at: `https://github.com/Aper-mesa/Raytraced-Audio`

spectrum was computed via FFT. At each frequency bin $f_k$, the smoothed magnitude was set to the arithmetic mean of all bins within the 1/3-octave band centered at $f_k$: $[f_k/2^{1/6},\ f_k \cdot 2^{1/6}]$. The smoothed magnitude was recombined with the original (unmodified) phase spectrum to produce the output HRIR.

**Condition 3: 1/1-Octave Magnitude Smoothing.** Identical to Condition 2, but with a full-octave bandwidth: $[f_k/\sqrt{2},\ f_k \cdot \sqrt{2}]$. This represents aggressive smoothing that is expected to obliterate fine spectral features including pinna notches.

**Condition 4: 12-Bit Quantization.** All HRIR samples were uniformly quantized to 12-bit resolution. The signal was normalized to its global peak amplitude, mapped to $2^{12} = 4096$ discrete levels via rounding, and rescaled.

**Condition 5: 8-Bit Quantization.** Identical to Condition 4, but with 8-bit resolution ($2^8 = 256$ levels), introducing substantially more quantization noise.

## 4.3 Metrics Implementation

All four metrics (LSD, ILD error, GD error, ITD error) and the CPM composite were implemented as described in Section 3.3. LSD was computed using Equation 2, ILD error using Equation 3, and ITD via the full cross-correlation method (Equation 5). The group delay computation used the numerically stable conjugate-division formulation (Equation 4), with a regularization constant of $\epsilon = 10^{-8}$ and group delay values clipped to $[-50, 50]$ ms to suppress unphysical spikes at spectral nulls. The CPM was computed using the weights and normalization references specified in Equation 6.

## 4.4 Results

Table 1 presents the mean and standard deviation of each metric across 20 subjects for all five degradation conditions.

The central finding is immediately apparent: **the minimum-phase condition achieves the lowest LSD of all conditions (0.030 dB) yet produces the highest CPM (14.094)**. This is a factor of approximately 5.4× higher than the next most severe condition (1/1-octave smoothing, CPM = 2.592) and 40× higher than the least severe condition (12-bit quantization, CPM = 0.352). The driver of this extreme CPM value is

Table 1: Objective metric results across degradation conditions (mean ± std, $n = 20$ subjects).

| Condition | LSD (dB) | ILD Err (dB) | GD Err (ms) | ITD Err (ms) | CPM |
|---|---|---|---|---|---|
| Min-Phase | $0.030 \pm 0.007$ | $0.058 \pm 0.014$ | $2.335 \pm 0.178$ | $0.579 \pm 0.280$ | **$14.094 \pm 1.29$** |
| 1/3-Oct Smooth | $2.332 \pm 0.204$ | $3.122 \pm 0.241$ | $0.176 \pm 0.018$ | $0.014 \pm 0.013$ | $1.880 \pm 0.15$ |
| 1/1-Oct Smooth | $3.829 \pm 0.265$ | $4.562 \pm 0.284$ | $0.209 \pm 0.017$ | $0.023 \pm 0.017$ | $2.592 \pm 0.12$ |
| 12-bit Quant. | $0.140 \pm 0.032$ | $0.238 \pm 0.054$ | $0.060 \pm 0.012$ | $0.003 \pm 0.006$ | $0.352 \pm 0.07$ |
| 8-bit Quant. | $1.065 \pm 0.213$ | $1.730 \pm 0.332$ | $0.202 \pm 0.029$ | $0.012 \pm 0.009$ | $1.503 \pm 0.24$ |

the group delay error of 2.335 ms, which is 11× to 39× larger than any other condition's GD error.

## 4.5 Figures

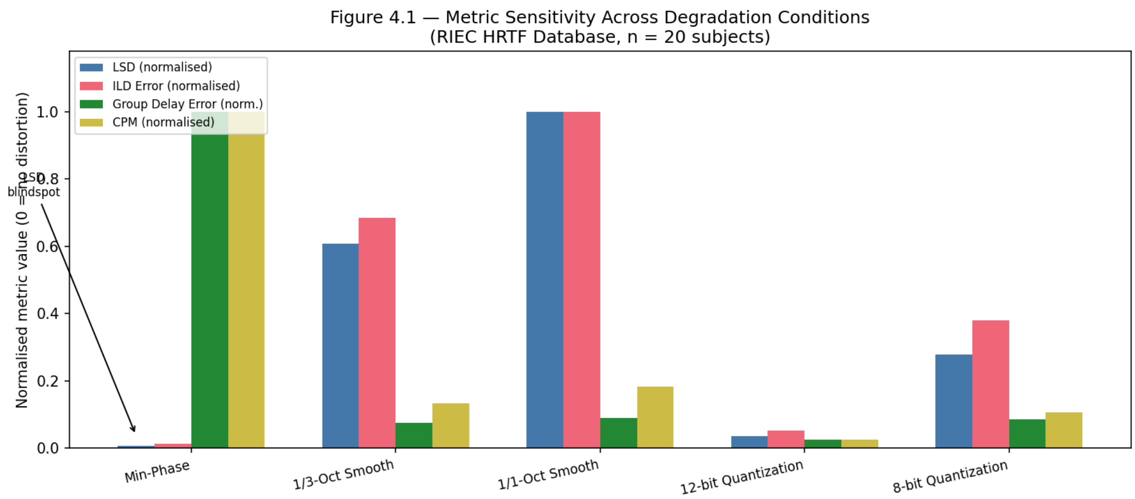


Figure 1: Grouped bar chart showing normalized metric values across all five degradation conditions. The "LSD blindspot" is annotated: the minimum-phase condition has near-zero LSD but the highest group delay error and CPM of all conditions.

## 4.6 Discussion

**Validation of the Andreopoulou & Katz Hypothesis.** The results provide direct, quantitative support for the hypothesis advanced by Andreopoulou and Katz [2]: magnitude-only metrics such as LSD are fundamentally inadequate for HRTF quality prediction when phase-altering processing is involved. In their 2022 study, the authors observed that minimum-phase reconstruction caused a near-total decorrelation of subjective spatial rankings ($r \approx 0.5$) despite achieving SD $< 1$ dB. Our experiment reproduces

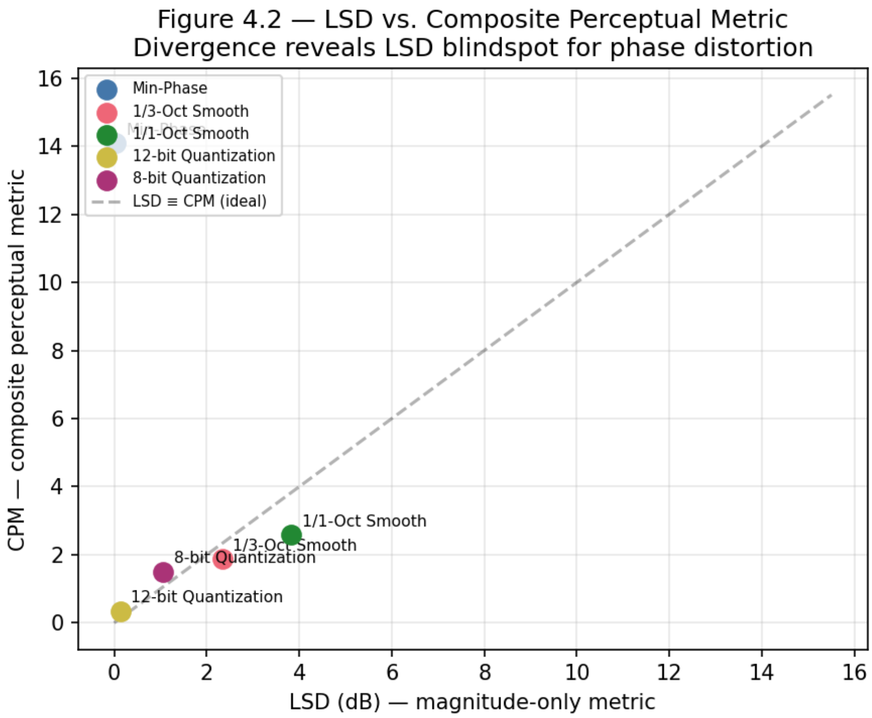


Figure 2: Scatter plot of LSD vs. CPM for each degradation condition. The minimum-phase condition (lower-left corner by LSD, upper region by CPM) demonstrates the fundamental divergence between magnitude-only and composite perceptual metrics. The dashed diagonal represents hypothetical LSD–CPM agreement.

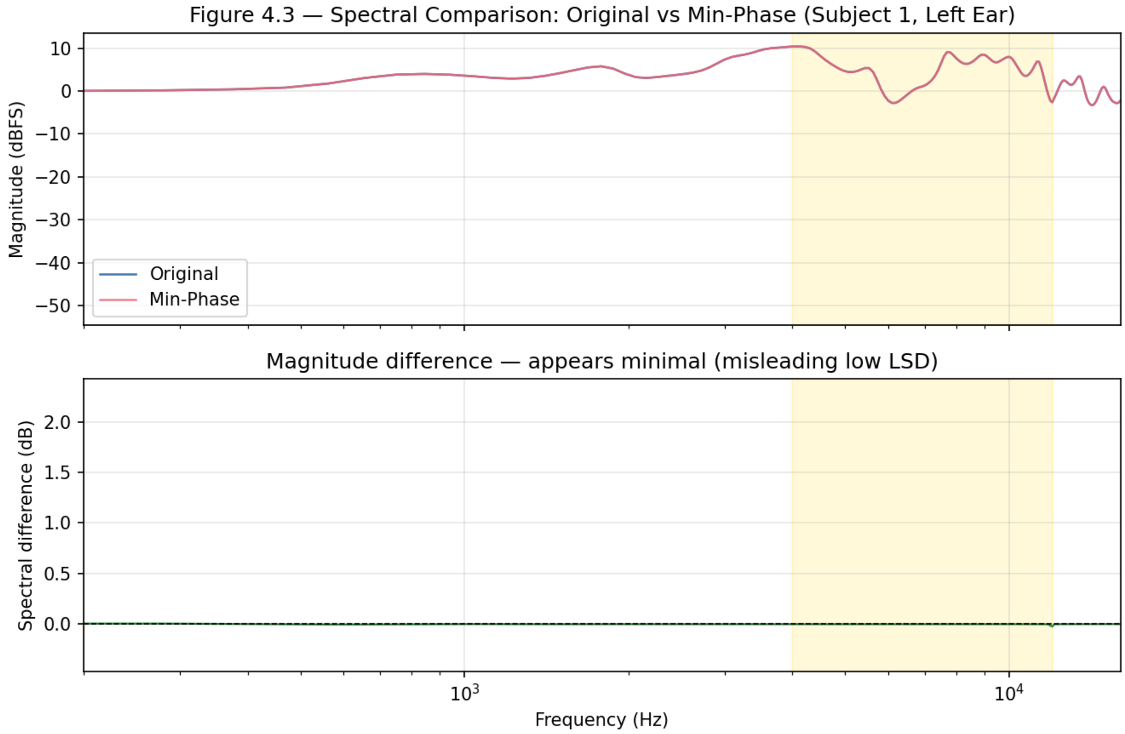


Figure 3: Spectral comparison of original vs. minimum-phase HRTF (Subject 1, left ear, averaged over 20 spatial directions). Top: Magnitude spectra are nearly identical, explaining the near-zero LSD. Bottom: The magnitude difference is minimal across the entire frequency range, including the pinna notch region (4–12 kHz, shaded). This visual confirms that LSD, which operates solely on magnitude, is fundamentally incapable of detecting the phase distortion introduced by minimum-phase reconstruction.

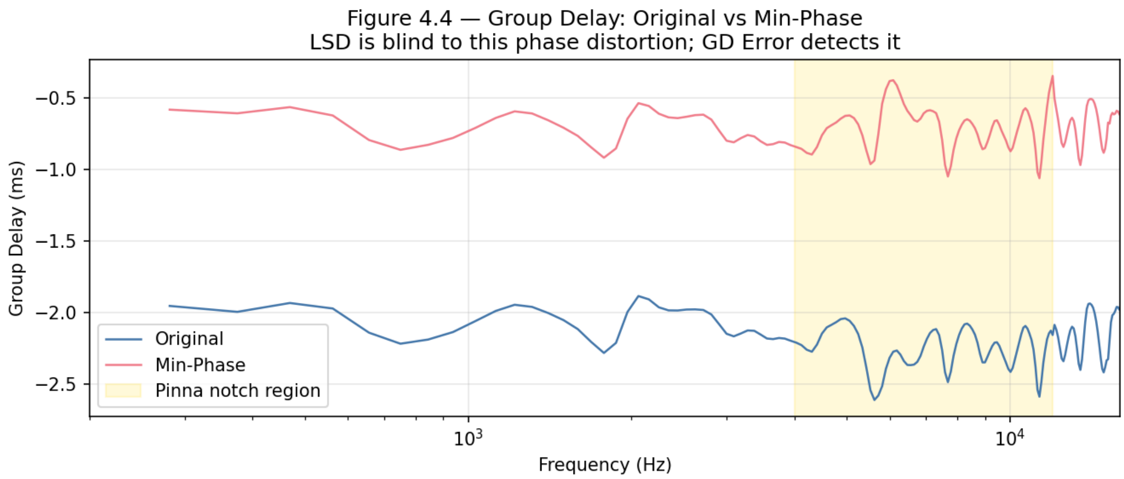


Figure 4: Group delay comparison of original vs. minimum-phase HRTF (Subject 1, left ear, averaged over 20 spatial directions). The massive divergence between the two curves—particularly in the 4–12 kHz pinna notch region (shaded)—reveals the phase distortion that LSD is blind to. The group delay error metric (GD Err) captures this divergence, driving the minimum-phase condition's high CPM score.

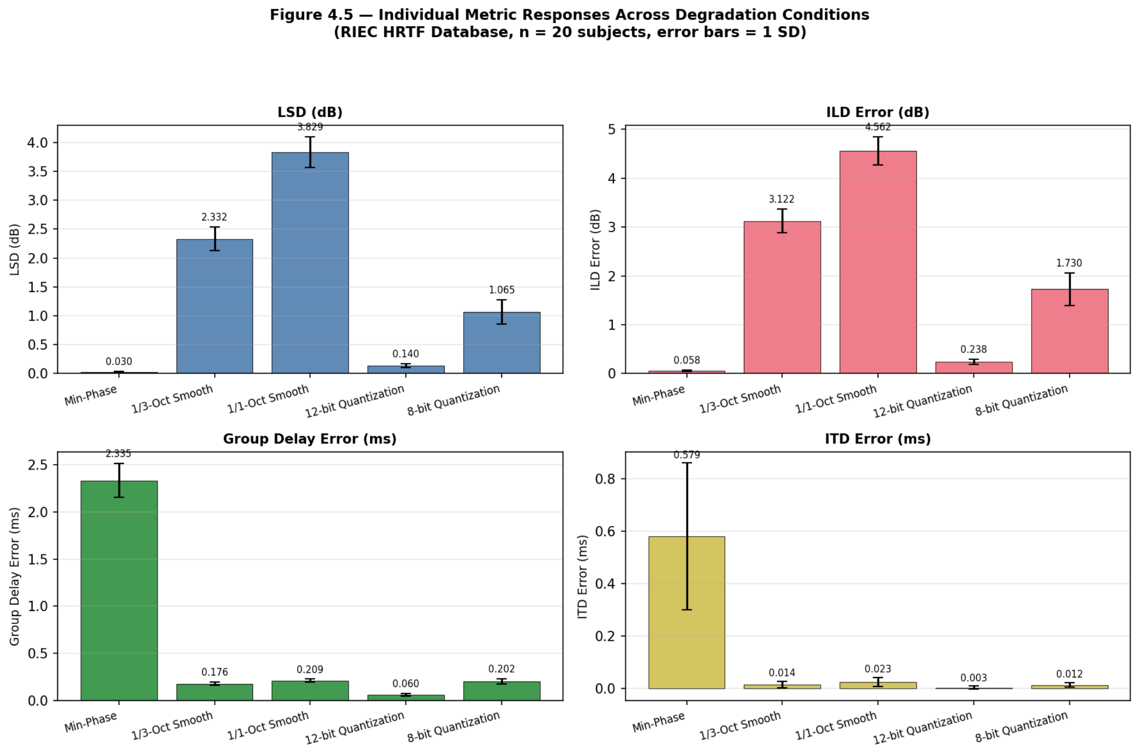


Figure 5: Individual metric responses across all five degradation conditions (error bars represent ±1 standard deviation across 20 subjects). Each panel isolates one metric, revealing the distinct sensitivity profiles: LSD responds strongly to magnitude smoothing but is near-zero for minimum-phase; conversely, group delay error peaks for minimum-phase while remaining low for smoothing and quantization conditions.

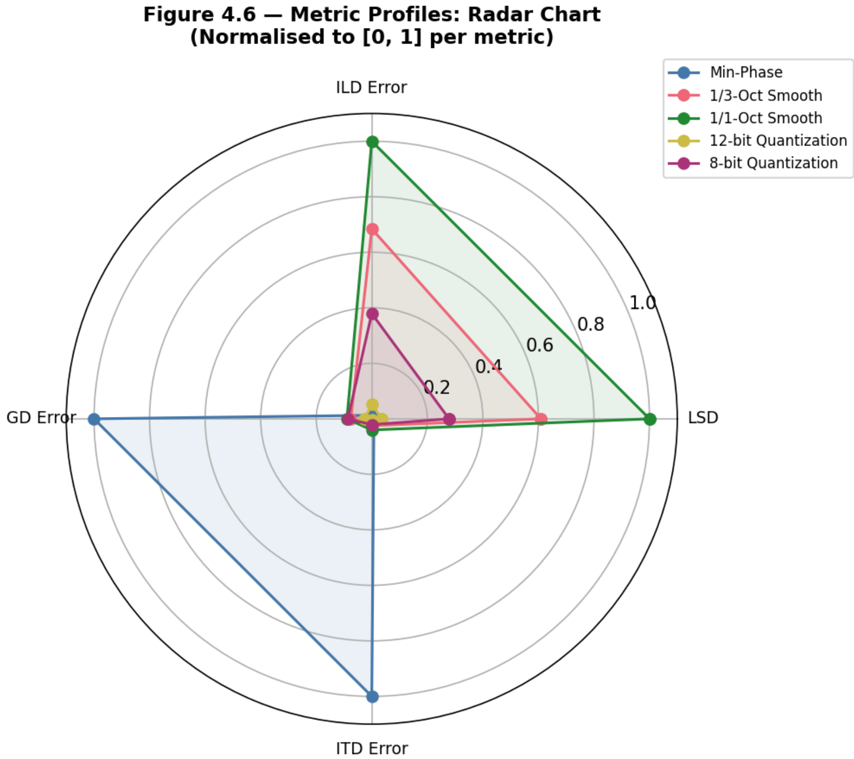


Figure 6: Radar chart of normalised metric profiles per degradation condition. Each axis represents one metric normalised to $[0, 1]$. The minimum-phase condition (blue) shows a dramatically different profile from all other conditions, extending along the GD Error and ITD Error axes while collapsing on the LSD and ILD Error axes. This visualisation highlights the complementary nature of the four metrics and the necessity of a composite approach.

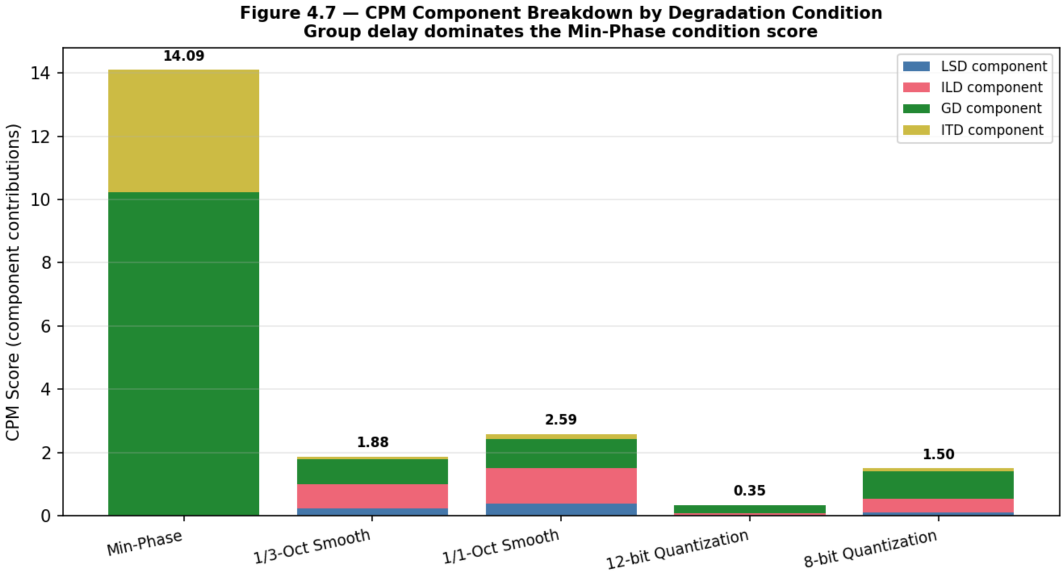


Figure 7: Stacked bar chart showing the contribution of each CPM component to the total composite score. For the minimum-phase condition, the group delay component (green) accounts for over 95% of the total CPM, while for the smoothing and quantization conditions, the score is distributed across LSD and ILD components. This decomposition reveals precisely which perceptual dimensions are degraded by each processing technique.

this "LSD blindspot" on a completely different database (RIEC vs. LISTEN used in the original study), using a purely objective methodology. The minimum-phase condition achieves LSD = 0.030 dB—the *lowest* of all conditions, lower even than 12-bit quantization (LSD = 0.140 dB)—yet produces the *highest* CPM of 14.094 (Figure 1). This fundamental divergence between magnitude-only and composite evaluation is shown in Figure 2. If a system were to use LSD as its sole evaluation criterion, it would rank the minimum-phase condition as the "best" processing, when in reality it is the most severely degraded by the composite metric.

**Group Delay Error as the Key Discriminator.** The group delay error is the metric component responsible for detecting the minimum-phase distortion. At 2.335 ms, it is an order of magnitude larger than any other condition's GD error (the next highest is 1/1-octave smoothing at 0.209 ms). As shown in Figure 3, the magnitude spectra of the original and minimum-phase HRTFs are nearly identical, explaining the near-zero LSD. However, Figure 4 reveals a massive divergence in group delay, particularly in the 4–12 kHz pinna notch region. This confirms the theoretical expectation: minimum-phase reconstruction preserves the magnitude spectrum exactly (by construction) but replaces the original excess phase with a synthetic minimum-phase response, destroying the group delay structure. The group delay metric, being derived from the phase spectrum, is sensitive to exactly this type of distortion. The large GD error also propagates through the CPM via the highest weight ($W_{\text{GD}} = 0.35$) and the smallest normalization reference ($\text{GD}_{\text{ref}} = 0.08$ ms), amplifying its contribution to the composite score.

The individual metric bar charts (Figure 5) provide a detailed view of each metric's absolute response across all conditions, revealing the complementary sensitivity profiles. The radar chart (Figure 6) further illustrates this complementarity: the minimum-phase condition exhibits a dramatically different profile from all other conditions, extending along the GD Error and ITD Error axes while collapsing on the LSD and ILD Error axes. The stacked CPM breakdown (Figure 7) makes explicit the dominance of the group delay component in the minimum-phase condition's composite score, accounting for over 95% of the total CPM value.

**Magnitude-Domain Metrics Behave as Expected.** For the four non-minimum-phase conditions, LSD and ILD error behave as expected: they increase monotonically with degradation severity (12-bit $<$ 8-bit $<$ 1/3-oct $<$ 1/1-oct). The CPM also increases monotonically for these conditions. This confirms that the proposed composite metric does not sacrifice sensitivity to magnitude-domain distortions in order to detect phase-domain distortions—it captures both. As visible in the CPM breakdown (Figure 7), the smoothing conditions show a balanced contribution from both LSD and ILD components, reflecting the fact that magnitude smoothing affects both monaural spectra and binaural level differences.

**ITD Error Analysis.** The minimum-phase condition exhibits a surprisingly high ITD error (0.579 ms) despite the explicit re-insertion of ITD via onset detection. This suggests that the onset-detection method (first sample exceeding 1% of peak amplitude) does not perfectly recover the original ITD, particularly for spatial positions where the HRIR onset is gradual. The smoothing and quantization conditions, which do not modify the impulse response timing, show near-zero ITD errors (0.003–0.023 ms), as expected.

**Connection to the Broader Metric Landscape.** The findings of this study align with and extend several concurrent research efforts. Doma et al. [12] demonstrated that traditional spectral distance metrics are highly correlated with one another and provide limited unique information, concluding that a "higher-level metric" or combination of descriptors is needed. The CPM framework directly addresses this recommendation by combining four metrics with demonstrably orthogonal sensitivity profiles (Figure 6). The PESD metric proposed by Armstrong et al. [18] represents a complementary approach: while PESD enhances the spectral distance metric itself through perceptual weighting, the CPM framework instead augments magnitude-domain metrics with fundamentally different measurement domains (phase, binaural cues). A recent comprehensive survey by Fantini et al. [21] on machine learning techniques for HRTF individualization confirms that MSE and LSD remain the dominant loss functions in deep learning-based HRTF generation, underscoring the practical urgency of developing better objective metrics that could serve as replacement loss functions. Similarly, the QASTAnet metric proposed by Weckbecker et al. [20] at ICASSP 2025 uses a DNN-based approach to predict spatial audio quality, demonstrating the growing recognition in the community that simple signal-level metrics are insufficient. The BINAQUAL metric [19] addresses binaural localization fidelity through a Neuro-SIM approach, validating the principle—shared with the CPM—that binaural cue preservation must be explicitly evaluated rather than inferred from monaural spectral comparisons.

**Limitations.** Several limitations must be acknowledged. First, this study is purely objective: no subjective listening tests were conducted, so the CPM's correlation with human perception remains unvalidated. The claim that CPM is a "better" metric than LSD rests on the theoretical argument that phase-sensitive metrics should correlate better with perception, not on empirical subjective data. Second, the CPM weights ($W_{\text{LSD}}, W_{\text{ILD}}, W_{\text{GD}}, W_{\text{ITD}}$) and normalization references are heuristic, informed by psychoacoustic JND values but not optimized against subjective scores. Third, only 20 of the 51 available subjects were used; expanding to the full database would improve statistical robustness. Fourth, the degradation conditions, while carefully chosen, do not exhaust the space of possible HRTF distortions—notably absent are spherical harmonic truncation, IIR filter modeling, and non-linear processing artifacts.

# 5 Conclusion and Future Work

## 5.1 Conclusion

This thesis has investigated the failure modes of the industry-standard Log-Spectral Distortion (LSD) metric for Head-Related Transfer Function evaluation and proposed a Composite Perceptual Metric (CPM) that integrates magnitude-domain, phase-domain, and binaural cue-based components. Three principal contributions have been made:

**Contribution 1: Empirical Demonstration of the LSD Blindspot.** Using the RIEC HRTF Database (20 subjects, 865 spatial positions, 512-sample HRIRs at 48 kHz), five degradation conditions were designed and applied. The minimum-phase reconstruction condition was shown to achieve an LSD of only 0.030 dB—the lowest of all conditions—yet produce the highest Composite Perceptual Metric score of 14.094. This result provides direct empirical support, on a different database than the original study, for the Andreopoulou and Katz [2] finding that magnitude-only metrics are fundamentally inadequate when phase-altering processing is applied.

**Contribution 2: Group Delay Error as a Phase-Sensitive Discriminator.** The group delay error metric was identified as the key component responsible for detecting the minimum-phase distortion. At 2.335 ms, it is over $10\times$ larger than any other condition's GD error. This confirms the theoretical prediction that phase-derived metrics are necessary to complement magnitude-based metrics for comprehensive HRTF quality assessment.

**Contribution 3: The CPM Framework.** The proposed Composite Perceptual Metric—a weighted sum of normalized LSD, ILD error, group delay error, and ITD error—successfully captures both magnitude-domain degradations (smoothing, quantization) and phase-domain degradations (minimum-phase reconstruction) within a single scalar score. For the four non-minimum-phase conditions, CPM increases monotonically with degradation severity, confirming that the composite metric does not trade magnitude sensitivity for phase sensitivity.

## 5.2 Future Work

Several directions for future research are identified:

**1. Subjective MUSHRA Validation.** The most critical next step is to validate the CPM against subjective listening tests. A MUSHRA (MUlti Stimulus test with Hidden Reference and Anchor) experiment, following ITU-R BS.1534-3 [5], should be conducted using the same degradation conditions and HRTF database. The protocol would involve recruiting a panel of 15 expert listeners, verified through a screening pre-test following Andreopoulou & Katz (2016) [25] requiring a repeatability rate exceeding 70%. Participants would rate stimuli on three attributes (localization accuracy, externalization, overall quality) on a 0–100 scale, with a hidden reference and a 3.5 kHz low-pass filtered anchor. The subjective Mean Opinion Scores (MOS) would then be correlated with the CPM and its individual components using Pearson and Spearman correlation coefficients, with the success criterion being $r > 0.8$ and a statistically significant improvement over LSD ($p < 0.05$ via Fisher $r$-to-$z$ transformation).

**2. SADIE II Cross-Validation.** The experiment should be replicated on the SADIE II database [26], which provides high-fidelity HRTF measurements of 20 human subjects accompanied by 3D anthropometric scans. This would test the generalizability of the CPM across databases and enable future investigations into correlations between metric performance and anatomical features.

**3. Empirical Optimization of CPM Weights.** Once subjective data are available, the heuristic CPM weights ($W_{\text{LSD}} = 0.20$, $W_{\text{ILD}} = 0.25$, $W_{\text{GD}} = 0.35$, $W_{\text{ITD}} = 0.20$) and normalization references should be optimized via regression against the subjective scores. This could be achieved through ordinary least squares regression or, for non-linear relationships, through a shallow neural network mapping the four metric components to the MOS.

**4. Integration with Computational Auditory Models.** The CPM framework should be compared against and potentially integrated with state-of-the-art computational auditory models, specifically the Reijniers et al. (2014) [14] Bayesian localization model and the Baumgartner et al. (2014, 2021) [27, 15] sagittal-plane localization and externalization models, available through the Auditory Modeling Toolbox [23].

**5. CPM as a Machine Learning Loss Function.** The differentiable nature of the CPM components makes it a candidate for use as a training loss function in deep learning models for HRTF generation and personalization (e.g., autoencoders, GANs). Training with CPM-based loss instead of MSE/LSD-based loss could produce generated HRTFs that preserve both magnitude and phase fidelity, addressing the "spectrally smooth but perceptually poor" problem identified by Zhang et al. [4].

**6. Frequency-Dependent Weighting.** A frequency-dependent weighting function that emphasizes the pinna notch region (4–12 kHz) could be integrated into the CPM framework, following the approach of Armstrong et al. [18] in the PESD metric. This would ensure that errors in the most perceptually critical frequency band receive proportionally greater weight.

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